



UNIVERSITY OF
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Superconductivity

Lecture 04 – “Going Critical”

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Previously

"Superconductivity is an **electromagnetic** phenomenon."

⇒ Maxwell's equations + experimental observation ($\rho = 0$ & $B = 0$)

$$\cdot \frac{\partial \mathbf{j}}{\partial t} = \frac{1}{\mu_0 \lambda_L^2} \mathbf{E},$$

$$\cdot \nabla \times \mathbf{j} = -\frac{1}{\mu_0 \lambda_L^2} \mathbf{B}.$$

"Superconductivity is a **thermodynamic** phenomenon."

⇒ Free energy, entropy, heat capacity ...

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Free energy

Finding conditions

Free energy = $f(T, P, B, \dots)$

"When does a material become a superconductor?"

⇒ "When does a material cease to be a superconductor?"

Superconductivity is destroyed if you do

- Raise temperature, $T > T_c$,
- Apply a strong field, $H > H_c$ or
- Flow large current (density), $j > j_c$.

Mapping a phase diagram

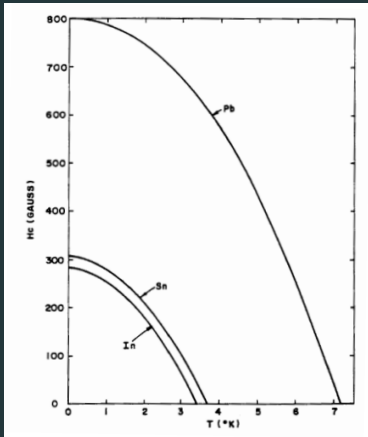


Figure 1: $H - T$ phase diagram of elemental metals, Pb, Sn and In.

What do we notice?

- T_c may link to H_c , and
- The curves may be universal.

Free energy (1/2)

"Superconductors do not like a field, apparently."

"Metals are a Pauli paramagnet having very small $\chi_P \sim 10^{-4}$."

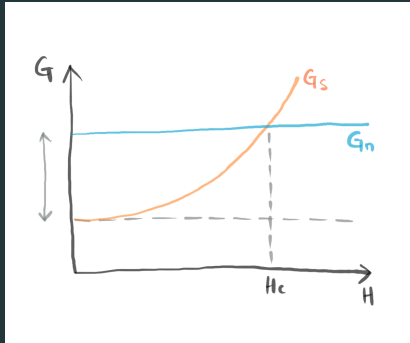


Figure 2: Gibbs free energy as a function of field, for normal metallic (G_n) and superconducting (G_s) states.

For a superconductor,
 $G_s(T, H) - G_s(T, 0) =$

$$\begin{aligned} &= \int_0^H dG \\ &= - \int_0^H \mu_0 \mathbf{M} \cdot d\mathbf{H} \\ &= \int_0^H \mu_0 H dH \\ &= \frac{\mu_0}{2} H^2. \end{aligned}$$

"Energy penalty" with a field.

Free energy (2/2)

For a metal,

$$\begin{aligned} G_n(T, H) - G_n(T, 0) &= - \int_0^H \mu_0 \mathbf{M} \cdot d\mathbf{H} \\ &= - \frac{\mu_0}{2} \chi_P H^2, \end{aligned}$$

"Energy gain" with a field.

Combining the above two for $H = H_c$, where $G_s = G_n$, gives

$$G_s(T, H_c) - G_s(T, 0) - [G_n(T, H_c) - G_n(T, 0)] = \frac{\mu_0}{2} H_c^2 + \frac{\mu_0}{2} \chi_P H_c^2, \quad (1)$$

which becomes,

$$G_s(T, 0) - G_n(T, 0) = - \frac{\mu_0}{2} (1 + \chi_P) H_c^2 \quad (2)$$

$$\simeq - \frac{\mu_0}{2} H_c^2, \quad (3)$$

Critical field

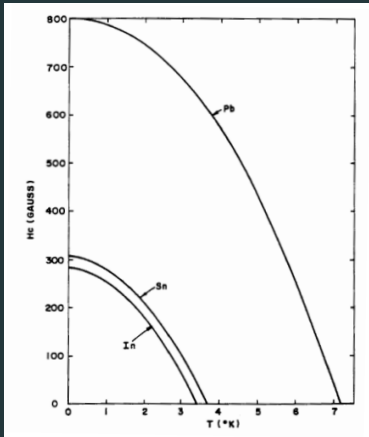


Figure 3: $H - T$ phase diagram of elemental metals, Pb, Sn and In.

$$G_s(T, 0) - G_n(T, 0) \simeq -\frac{\mu_0}{2} H_c^2$$

We are ready to calculate entropy, heat capacity, ...

$$S = -\frac{dG}{dT},$$

$$C = T \frac{dS}{dT}.$$

Entropy

Thermodynamic functions

$$S = - \frac{dG}{dT},$$
$$C = T \frac{dS}{dT}.$$

One can measure C in laboratory, then integrating C/T with temperature would give S .

Entropy (1/4)

We will use the following expression,

$$G_s(T, H_a) - G_n(T, H_a) \simeq -\frac{\mu_0}{2} (H_c^2 - H_a^2)$$

Let us first calculate the entropy for a superconducting state.

$$\begin{aligned} S_s(T, H_a) &= -\frac{dG_s(T, H_a)}{dT} \\ &= -\frac{d}{dT} \left[G_n(T, H_a) - \frac{\mu_0}{2} (H_c^2 - H_a^2) \right] \\ &= -\frac{dG_n(T, 0)}{dT} + \frac{d}{dT} \left(\frac{\mu_0}{2} H_c^2 \right) - \frac{d}{dT} \left(\frac{\mu_0}{2} H_a^2 \right) \\ &= S_n(T, 0) + \mu_0 H_c \left(\frac{dH_c}{dT} \right). \end{aligned}$$

Entropy (2/4)

$$S_s(T, H_a) = S_n(T, 0) + \mu_0 H_c \left(\frac{dH_c}{dT} \right).$$

What does this tell us?

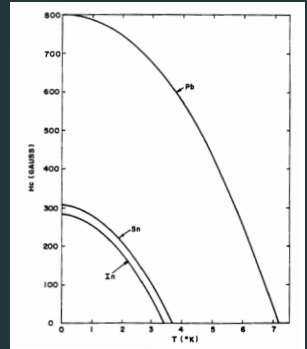


Figure 4: $H - T$ phase diagram of elemental metals, Pb, Sn and In.

Entropy (3/4)

$$S_s(T, H_a) = S_n(T, 0) + \mu_0 H_c \left(\frac{dH_c}{dT} \right).$$

"A superconducting state is more ordered than in normal state."

$$S_s \leq S_n$$

since from experiments

$$\frac{dH_c}{dT} \leq 0.$$

What would you expect further for $T = T_c$ or $T = 0$?

- $H_c(T = T_c) = 0 \rightarrow S_s = S_n$ at $T = T_c$
- The 3rd law of thermodynamics $\rightarrow S_s = S_n$ at $T = 0$

Note that the latter leads to $dH_c/dT|_{T=0} = 0$ consistent with the experimental observation.

Entropy (4/4)

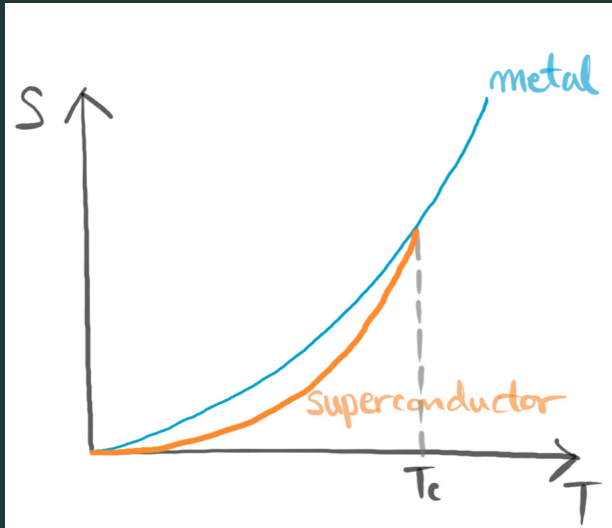


Figure 5: Entropy as a function of temperature.

Heat capacity

Heat capacity from entropy

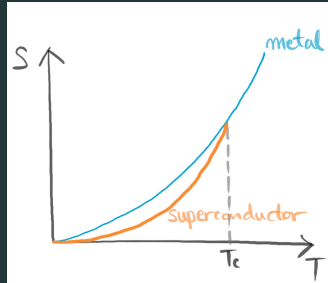


Figure 6: Entropy as a function of temperature.

Guess the curve form for heat capacity from¹

$$C = T \frac{dS}{dT}$$

¹We discuss only the electronic contribution to heat capacity, tentatively excluding the contribution from lattice which is presumed to remain the same between normal and superconducting states.

Heat capacity from entropy (1/3)

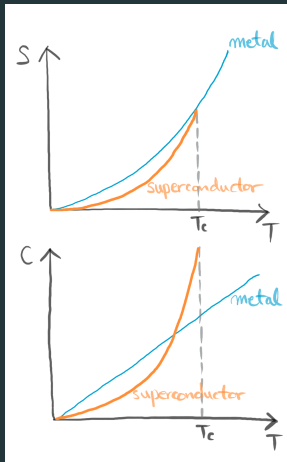


Figure 7: Entropy (upper) and heat capacity (lower) as a function of temperature.

Heat capacity from entropy (2/3)

$$\frac{C_s}{T} - \frac{C_n}{T} = \frac{d}{dT} (S_s - S_n) \quad (4)$$

$$= \frac{d}{dT} \left(\mu_0 H_c \frac{dH_c}{dT} \right) \quad (5)$$

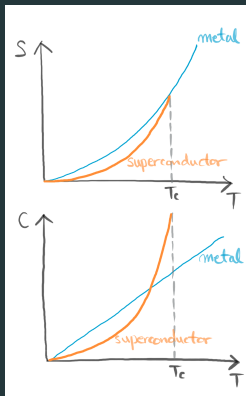
$$= \mu_0 \left(\frac{dH_c}{dT} \right)^2 + \mu_0 H_c \frac{d^2 H_c}{dT^2}. \quad (6)$$

$$C_s(T_c) - C_n(T_c) = \mu_0 T_c \left(\frac{dH_c}{dT} \right)^2. \quad (7)$$

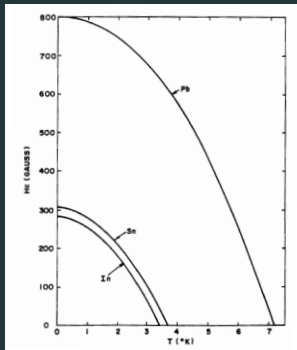
Does the above agree with our expectation?

Heat capacity from entropy (3/3)

$$C_s(T_c) - C_n(T_c) = \mu_0 T_c \left(\frac{dH_c}{dT} \right)^2. \quad (8)$$



Some inputs from experiments (1/3)



To make a further step from using

$$C_S(T_c) - C_n(T_c) = \mu_0 T_c \left(\frac{dH_c}{dT} \right)^2,$$

one may use the empirical form

$$H_c(T) = H_c(0) \left[1 - \left(\frac{T}{T_c} \right)^2 \right].$$

For instance, one can obtain

$$\begin{aligned} C_S(T_c) - C_n(T_c) &= 4T_c \mu_0 \left(\frac{H_c(0)}{T_c} \right)^2 \\ &\equiv 2\gamma T_c. \end{aligned}$$

Some inputs from experiments (2/3)

We could have started using

$$H_c(T) = H_c(0) \left[1 - \left(\frac{T}{T_c} \right)^2 \right].$$

from the beginning, to obtain

$$S_s - S_n = -2\mu_0 \left(\frac{H_c(0)}{T_c} \right)^2 \left[1 - \left(\frac{T}{T_c} \right)^2 \right] T,$$

and

$$\frac{C_s}{T} - \frac{C_n}{T} = -2\mu_0 \left(\frac{H_c(0)}{T_c} \right)^2 \left[1 - 3 \left(\frac{T}{T_c} \right)^2 \right],$$

(this is left for your as an exercise)

Some inputs from experiments (3/3)

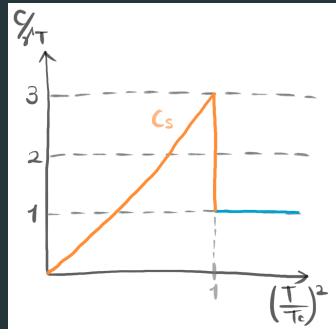
$$\frac{C_s}{T} - \frac{C_n}{T} = -2\mu_0 \left(\frac{H_c(0)}{T_c} \right)^2 \left[1 - 3 \left(\frac{T}{T_c} \right)^2 \right],$$

Using our knowledge that C for a metal is linear in temperature, i.e., $C_n/T \rightarrow \text{const.}$ as $T \rightarrow 0$,

$$\begin{aligned} \frac{C_n}{T} &= 2\mu_0 \left(\frac{H_c(0)}{T_c} \right)^2 \\ &= \gamma \end{aligned}$$

and

$$\begin{aligned} \frac{C_s}{T} &= 3 \times 2\mu_0 \left(\frac{H_c(0)}{T_c} \right)^2 \left(\frac{T}{T_c} \right)^2 \\ &= 3\gamma \left(\frac{T}{T_c} \right)^2. \end{aligned}$$



Experiments conclude

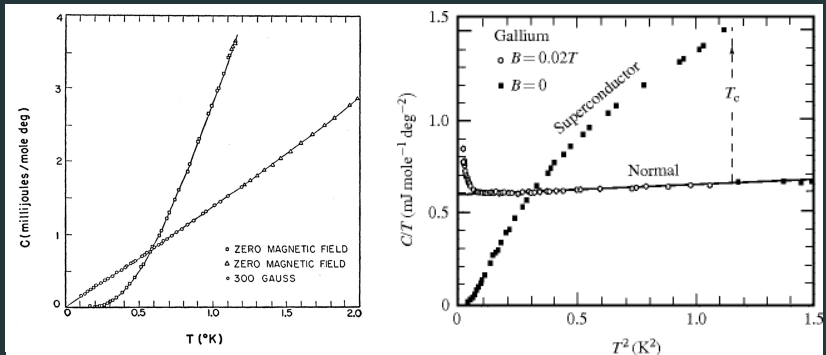


Figure 8: Temperature dependence of specific heat of Al (left, N. E. Phillips, Phys. Rev. B **114**, 676, 1959) and Ga (right, C. Kittel, Introduction to Solid State Physics, John Wiley & Sons, 1996) in zero and finite fields.

Critical currents

Critical currents (1/2)

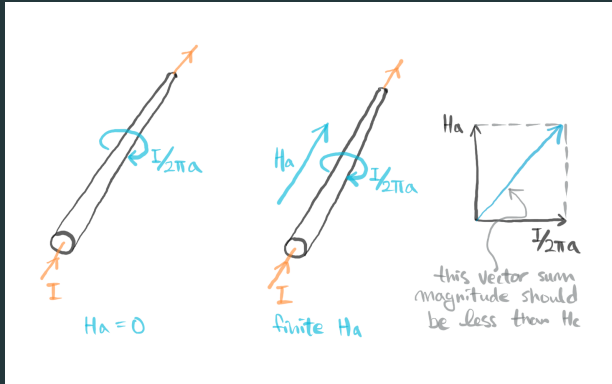


Figure 9: Currents and fields for a superconducting wire.

Critical currents (2/2)

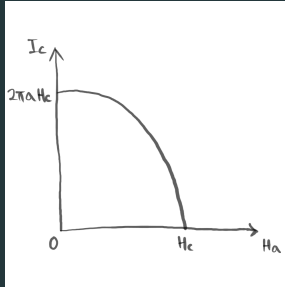


Figure 10: Critical current vs critical field diagram.